Permutation Patterns

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Outline

Combinatorics

Combinatorics is concerned with the study of...

- arrangements
- patterns
- designs
- assignments
- schedules
- connections
- configurations

Combinatorical Problems

- A computer scientist considers *patterns* of digits and switches to encode complicated statements.
- A shop supervisor prepares *assignments* of workers to tools or to work areas.
- An agronomist *assigns* test crops to particular fields.
- An electrical engineer considers alternative *configurations* for a circuit.
- An industrial engineer considers alternative production *schedules* and workplace *configurations* to maximize efficient production.
- A university scheduling officer *arranges* class meeting times and students' *schedules*.
- A chemist considers possible *connections* between various atoms and molecules and *arrangements* of atoms into molecules.
- A transportation officer *arranges* bus or plane *schedules*.
- A linguist considers *arragnements* of words in unknown alphabets.

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Counting: How many arrangements are there?

Optimization: Among all possible arrangements, what is the best one according to some criterion?

Basic Counting Rules

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$$5 + 4 = 9$$

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$$5 \cdot 4 = 20$$

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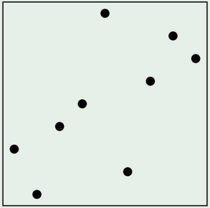
- S_n is the set of all permutations of length n.
- Common to plot permutations graphically by points $(i, \sigma(i))$ where $\sigma = \sigma(1) \ldots \sigma(n)$.

Permutation

Example

Consider $\sigma = 314592687 \in S_9$.



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$$|S_n| = n(n-1)(n-2)\cdots(n-i)\cdots 1 = n!$$

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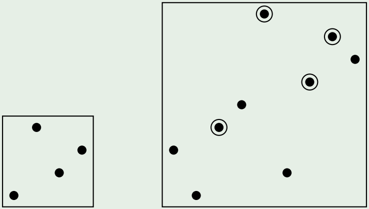
Example

$1423 \leq 314592687$ because 4968 (among others) is ordered the same way:



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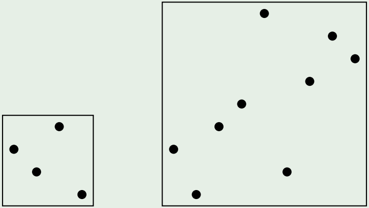
Example

$314592687 \in S_9(3241)$ because $3241 \not\leq 314592687$:



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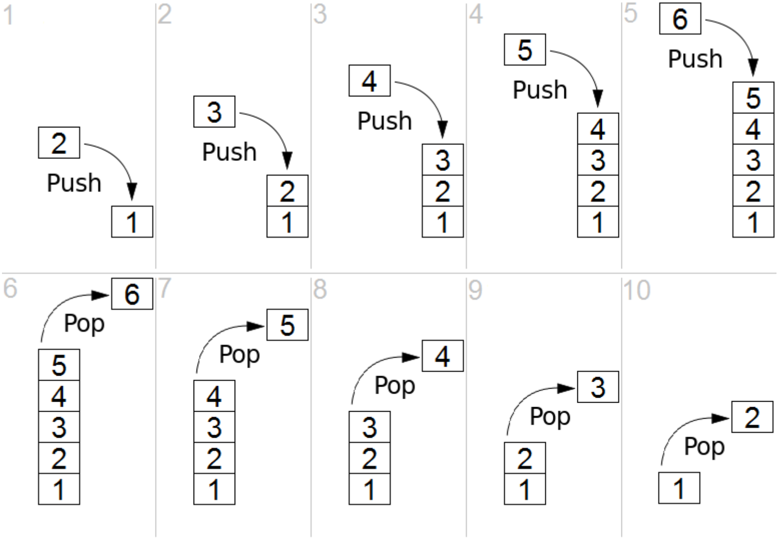
Stack

- Linear structure with insertion and deletion on same end
- "Push" and "pop"
- Last in, first out (LIFO)
- E.g., stack of plates in cafeteria

Stack Example



Sorting with a Stack

Knuth's algorithm

- Initialize an empty stack
- For each input value x:
  - While the stack is nonempty and x is larger than the top item on the stack, pop the stack to the output
  - Push x onto the stack
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Example (4132)

1. Push 4 onto stack
2. Push 1 onto stack
3. Pop 1 off of stack
4. Push 3 onto stack
5. Push 2 onto stack
6. Pop 2 off of stack
7. Pop 3 off of stack
8. Pop 4 off of stack

Sorting with a Stack

Example (3142)

1. Push 3 onto stack
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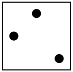

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A permutation σ is stack sortable if and only if $\sigma \in S_n(231)$.

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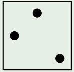
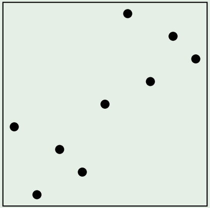
The permutation 413259687 is stack sortable.





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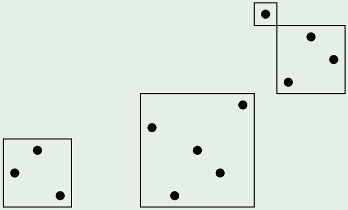

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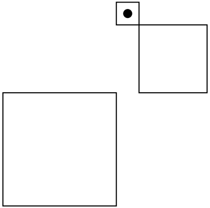

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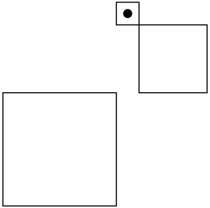

If $|S_n(231)| = c_n$, then $c_n = \sum_{i=1}^{n} c_{i-1} c_{n-i}$.

Counting Stack Sortable Permutations

Using some standard methods, we find

$$c_n = \frac{1}{n+1}\binom{2n}{n} = \frac{(2n)!}{(n+1)!n!} = C_n,$$

the *Catalan numbers*, which count many combinatorial objects:

- Wikipedia
- The Online Encylopedia of Integer Sequences